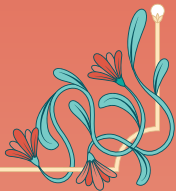


Beyond Immutable Machines: Introducing Cluster API In-Place Updates

Lennart Jern

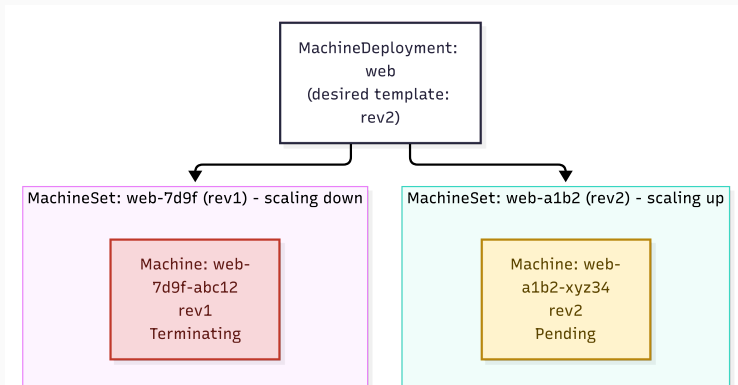
Ericsson Software Technology

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Background

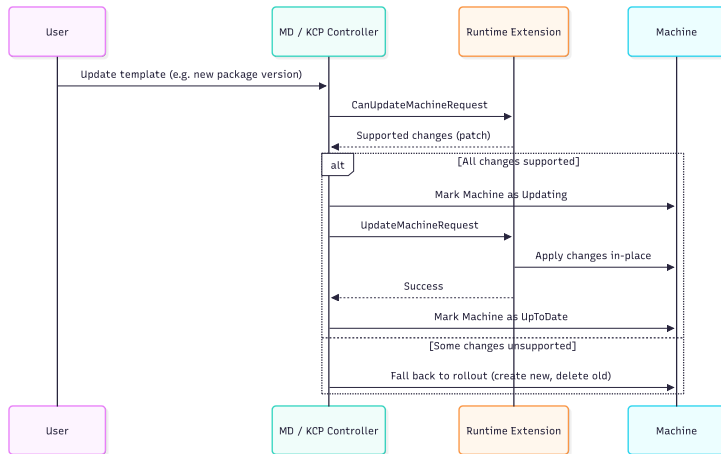
- Cluster API traditionally works with immutable Machines, similar to how Deployments and Pods work in Kubernetes.
- Changes are generally rolled out by replacing Machines.
- This approach is not going anywhere and has many benefits, but there is now an alternative: In-place updates.



Why In-place Updates?

- Replacing Machines can be disruptive, slow, and resource-intensive.
- Bare metal environments are particularly affected by this: Reboot, clean disks, re-install OS, just to then re-join the same cluster with a slightly changed configuration.
- Workloads sensitive to “node churn”
- Operating Systems built to be immutable and managed in a different way (e.g., CoreOS, Flatcar, Talos) become wasted effort when the OS is re-installed just to change a configuration.

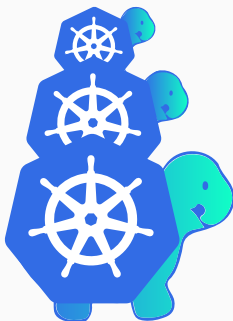
How In-place Updates Work ¹



¹<https://cluster-api.sigs.k8s.io/tasks/experimental-features/runtime-sdk/implement-in-place-update-hooks.html>

Conclusion

- Cluster API Machines can now be updated in-place by using Runtime Extensions.
- The feature is available in Cluster API v1.12.0 and later.²
- This enables specialized workflows for certain workloads and environments, improving efficiency and reducing disruptions.



²<https://kubernetes.io/blog/2026/01/27/cluster-api-v1-12-release/>

Thank you!

